

Assignment -2

Q1.What is a conditional statement in programming?

Ans. A statement used to make decisions in a program based on whether a condition is true or false.

Example: if-else, switch etc.

Q2. What does an if statement do?

Ans. if statement check the condition and runs code only if the condition is true.

Q3.What is the purpose of the else block?

Ans. when condition is false . it means condition is not true.

Q4. When is an else block executed?

Ans. An else block is executed when if condition is false.

Q5.what is a Boolean expression?

Ans. that return only true or false.

Q6. Write the syntax of a simple if statement?

Ans. if(condition)

```
{ // statements }
```

Q7.What operator is commonly used to test equality?

Ans. “==” operator is commonly used to test equality.

Q8. What does the modulus(%) operator return?

Ans. modulus operator always return Remainder.

Q9. How can you check if a number is even?

Ans. if a number is even by seeing if it is divisible by 2.

Condition : $\text{number} \% 2 == 0$ ----- even number.

Q10. How can you check if a number is odd?

Ans. if the number is not divisible by 2.

Condition : $\text{number} \% 2 != 0$ ----- odd number .

Q11. What is the purpose of else-if?

Ans. The purpose of else if is to check multiple conditions one by one when the first if condition is false.

Q12. What is a nested if statement?

Ans. nested if statement means if statement placed inside another if statement.

Q13. Can an if statement exist without an else?

Ans. yes , if statement exist without an else.

Q14. What is the output when a false condition is tested in an if statement without else?

Ans. The compiler (C/C++) simply skips the if block and moves to the next line of code when condition is false.

Q15. What is the role of braces{} in C/C++ conditionals ?

Ans. in C and C++ conditionals, braces{} are used to group multiple statements into a single block.

Q16. What does > mean?

Ans. the symbol > means “greater than”

It is used to compare two values.

Q17. What does < mean?

Ans. the symbol < means “less than “

It is also used to compare two values.

Q18. What does >= mean?

Ans. “>= “ means greater than or equal to

It checks if the left value is bigger than or equal to the right value.

Q19. What does <= means?

Ans. “<=” means less than or equal to

It checks if the right value is bigger than or equal to the left value.

Q20. What does != mean?

Ans. "!=" means not equal to .

It is a comparison operator used to check if two values are different.

Q21. What is a switch statement?

Ans. A switch statement is a control statement . it checks a value and runs the matching case.

***Q22 . What is the case label in a switch statement?**

Ans. A case label is a value in a switch statement that determines which block of code will execute.

Q23. Why is break used in switch?

Ans. Break is used in a switch statement to stop execution after a matching case is executed.

Q24. What happens if break is omitted?

Ans. if break is omitted in a switch statement , the program does NOT stop after the matching case.

Q25. What is the default case in switch?

Ans. The default case in a switch statement runs when none of the cases match.

Q26. Can switch compare string in C++?

Ans. switch statement cannot directly compare strings.

Q27. Which statement is better for many constant choices. If-else or switch?

Ans. for many constant choices, the switch statement is better than if-else.

Because it's code cleaner and easier to read and faster for constant values.

Q28. How do you test whether a value is positive?

Ans. firstly check if it is greater than zero.

Condition: `number > 0` ----- positive.

Q29. How do you test whether a value is negative?

Ans. to test whether a value is negative , you check if it is less than zero.

Condition : `number < 0` ----- negative .

Q30. How do you test whether a value is Zero ?

Ans. when `number == 0`

`If (number == 0)`

`{`

```
Printf("zero");  
}
```

Q31. Write a condition to check if age is at least 18 ?

Ans. the condition to check if age at least 18

```
if(age>=18)
```

Q32. Write a condition to check if marks are above 50?

Ans. if(marks>50)

Q33. What is decision making in programming?

Ans. Decision making programming means The program checks a condition and then decides what to do next.

Program choosing between options using conditions(if,else,switch).

Q34. Can nested if statements have multiple levels?

Ans. yes nested if statements ave multiple levels.

Example:

```
If(a>0){
```

```
If(b>0){
```

```
If(c>0){
```

```
Printf("All positive");
```

```
}  
}  
  
}
```

Q35. What is logical And (&&)?

Ans. Logical And (&&) is an operator used in programming to combine two conditions.

It return True only when both conditions are true

Example:

```
If(age>18 && age<40){  
    Printf("Eligible");}
```

It means here, both conditions must be true.

- age>18
- age<40

Eligible for exam.

Q36. What is logical OR (||)?

Ans . it is also used in programming to combine two condition but it return True when one condition is true or both condition are true .

Q37. What is logical NOT(!)?

Ans. Logical NOT (!) is an operator that reverses a condition.

If a condition is true, ! makes it false.

If a condition is false, ! makes it true.

Q38. How do you check if a year is divisible by 4?

Ans. condition:

```
if(year%4==0)
{
printf("year is divisible by 4");
}
else
{
printf("Not divisible by 4);
}
```

Q39.What is a leap year?

Ans. A leap year is a year that has 366 days instead of 365 days.

Ex.= 2024,2028,2000,1940 etc.

Q40.What is the first condition for a leap year?

Ans. first condition is (year%4==0)

Q41. How do you find the largest of two numbers?

Ans. I will apply condition between two numbers.

```
#include<iostream>
```

```
using namespace std;
```



```
int main()
{
    int float a , b;
    cout<<"Enter two numbers";
    cin>> a>> b;
    if(a>b)
    {
        printf("a is greater");
    }
    else
    {
        printf("b is greater");
    }
    return 0;
}
```

OUTPUT

Enter two numbers 45 67

b is greater

Q42. How do you find the largest of three number ?

Ans. to find the largest of three numbers, compare them using if-else or if or ternary operator.

C++ Program:

```

#include<iostream>

using namespace std;

int main()
{
int a, b, c;

cout<<"Enter three numbers;

cin>>a>>b>>c;

if(a>=b&& a>=c) cout<<"a is greater";

else if(b>=a && b>=c) cout<<"b is greater";

else

cout<<"Largest number is c";

return 0;

}

```

Example :

Input: 10 23 45

Output :

Largest number is c

Q43.What is an if-else –if ladder?

Ans. it's used when need to test multiple conditions one after another.]

Means the first condition is false, the next condition is checked, and so on. If none of the conditions are true, the else block executes.

Syntex:

(C programming):

```
if(condition 1)
```

```
{
```

```
Statement 1
```

```
}
```

```
else if (condition 2)
```

```
{
```

```
Statements 2
```

```
}
```

```
else if (condition 3)
```

```
{
```

```
Statement 3
```

```
}
```

```
else
```

```
{
```

```
Statement 4....
```

```
}
```

Q44. What is the difference between if and switch?

Ans.

IF	SWITCH
1. used to check conditions	1.used to select one option from multiple fixed values.
2. can handle complex conditions.	2.can only compare a variable with constant case values.

3. Uses if,else if,and else.	3.uses switch,case and default.
4. Can check ranges(e.g. marks>50)	4.can't directly check ranges

***Q45. Can switch use ranges directly?**

Ans. A switch statement cannot directly check ranges of values. For range-based conditions, if-else statement should be used.

Q46. What is fall –through in switch?

Ans. Fall-through in switch is a situation where, in the absence of a break statement , execution continues from the matched case to all subsequent cases.

Q47.What data type is usually used for conditions?

Ans. in C programming , the int data type is usually used for conditions, where 0 represents false and any non –zero value represents true. I modern C, the bool type can also be used.

Traditional C int.

Modern C bool.

Q48. Why are conditional statements important?

Ans.

1. Decision making = they allow the program to choose between different actions.
2. Control Flow = they control the flow of execution based on conditions.
3. Problem solving – used in real-world applications such as grading systems, ATM machines, and login systems.
4. Flexibility – Makes programs more dynamic and responsive to user input.

Q49. Give one real-life example of a conditional statement.

Ans. Traffic Signal

- If the traffic light is green, vehicles can move.
- If the traffic light is red, vehicles must stop.

Q50. Name four conditional structures covered?

Ans.

1. Simple if statement.
2. If-else statement.
3. If-else-if Ladder.
4. Switch statement.